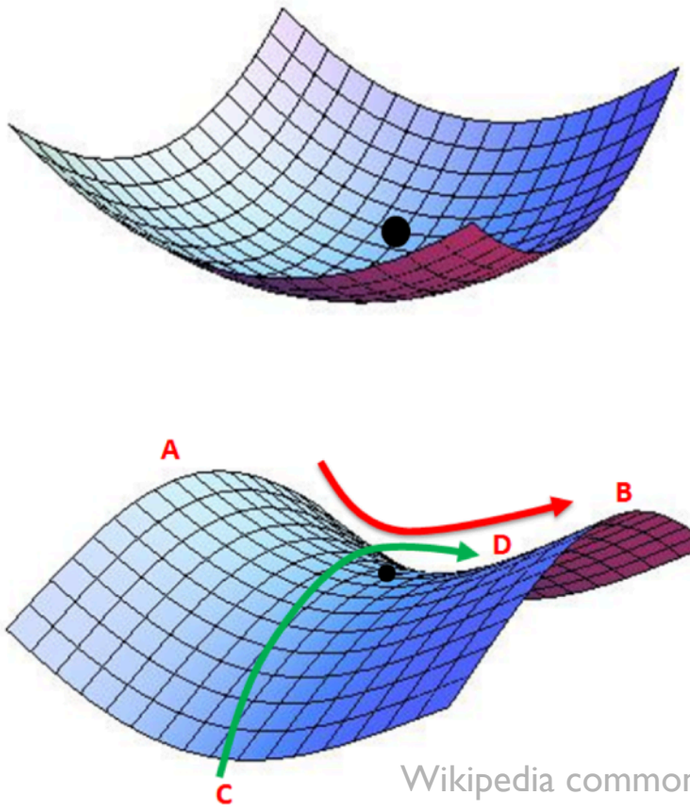
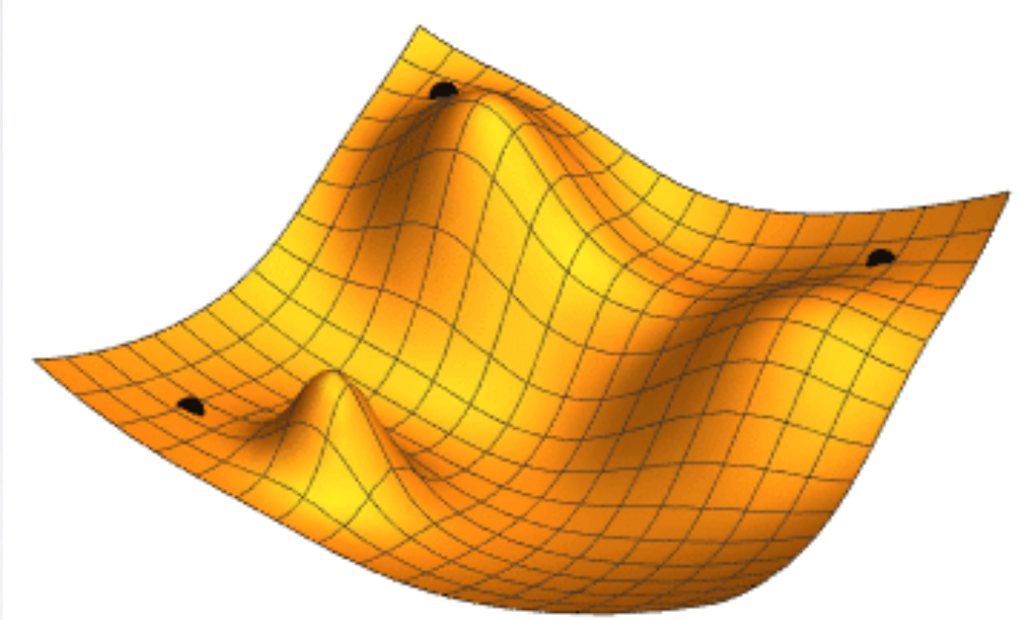
Week 7 - #2

Vector Analysis (II)



Wikipedia commons



gradient descent, Wikipedia commons

Today: Ch 6

Next Class: Ch 10.5, 10.7-10.9

Ji-hoon Kim (Seoul National University)

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Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

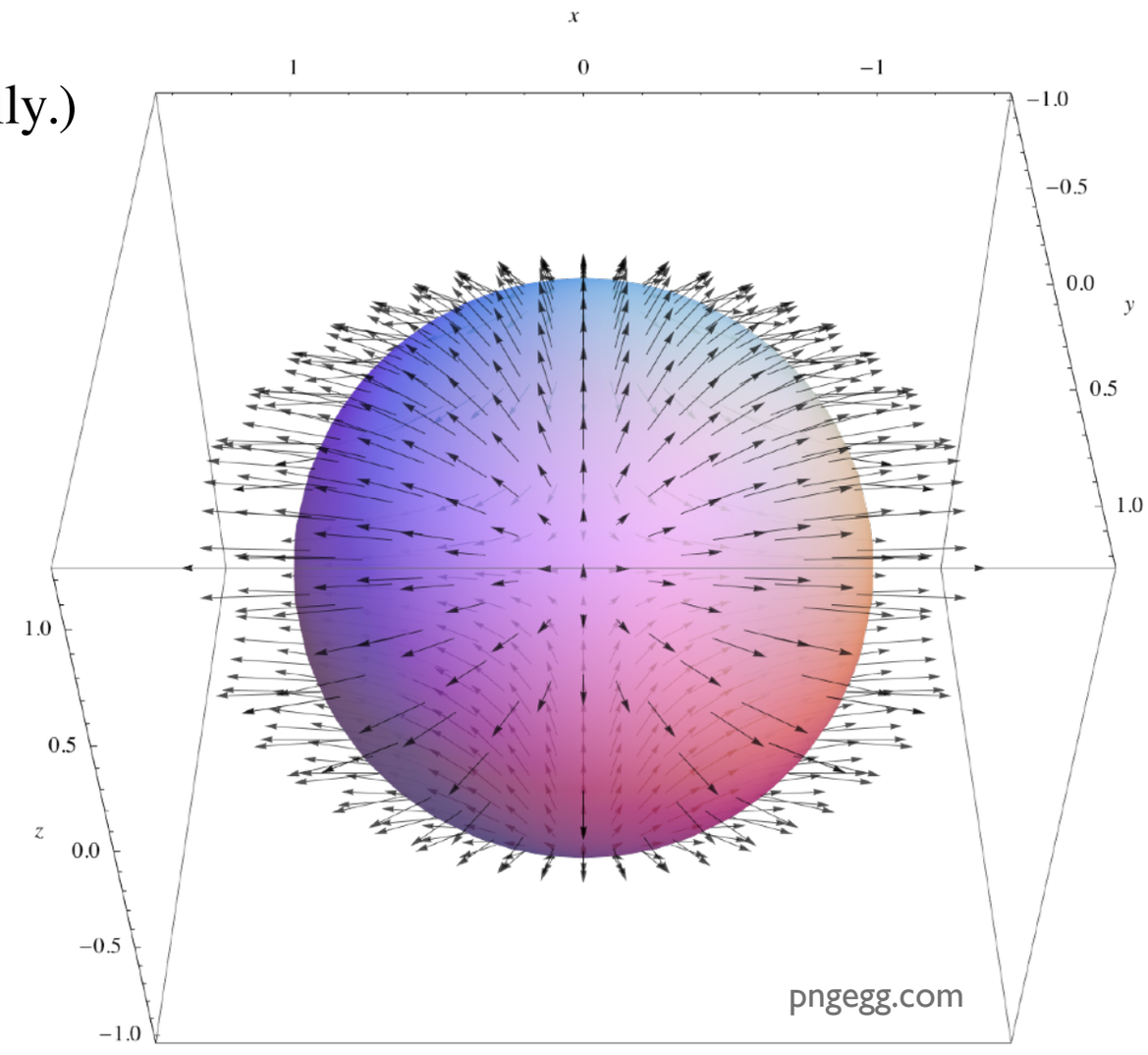
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #10

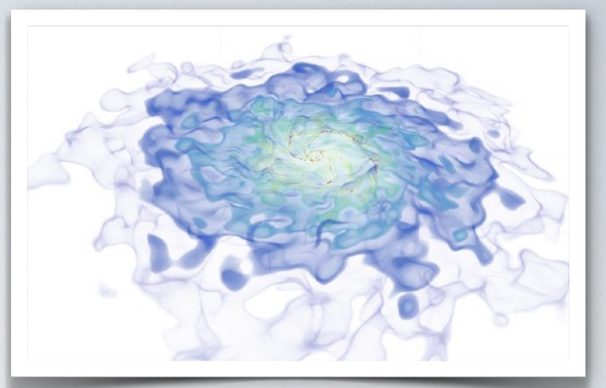
— [open book and open note, but no cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Explain, in layman's terms without detailed math or proofs, what the divergence theorem means. You may use a figure like the one below. Refer to your textbook for inspirations.

2. Boas, Chapter 6, Section 10, Problem 3
(Read the instruction at the top of the page carefully.)





No Class next Wednesday (Oct. 21)

Midterm Examination: October 24

- Oct. 24 (Sat), 19:30-21:00 pm
- 56-105 (student ID ending with odd number) or 106 (even number)
- Covers everything discussed in the class before the exam including Boas Chapters 1-6 and parts of Chapter 10.



msutoday.msu.edu

Mathematical Physics I (Fall 2026): Midterm Examination

Oct. 24, 2026

[total 20 pts, closed book/cellphone, no calculator, 90 minutes]

• First, make sure you have all 6 answer sheets. Write down your name and student ID on each of all 6 answer sheets. Then, number the sheets from (1) to (6) on the top right corner. Your answer to each problem must *only* be in the sheet with the matching number (e.g., your answer to Problem 2 must *only* be in sheet (2)). After the exam, you will separately turn in all 6 answer sheets, even if some sheets are still blank.

• Make sure you have all 6 problems. Have a quick look through them all and portion your time wisely. If you find any issue or question, you *must* raise it in the first 30 minutes. You have to stay in the room for that 30 minutes even if you have nothing to write down.

• Make your writing easy to read. Illegible answers will *not* be graded.

Midterm: On The Day of the Exam (I)

- Arrive by no later than 19:15pm. One of the first floor doors to Bldg 56 will be propped open between 19:00-19:30pm.

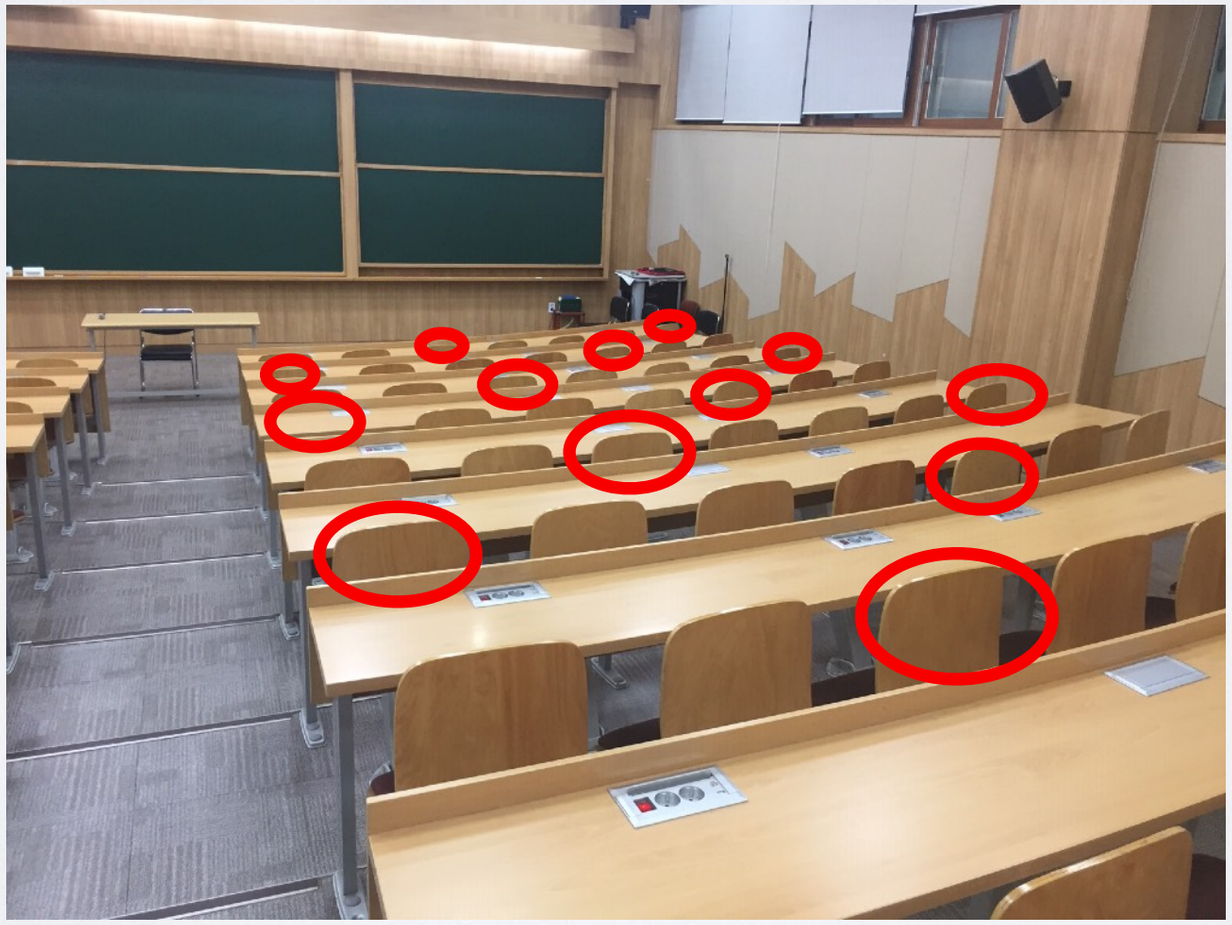
- If you have a fever or a cough, or have had any symptoms, you must report to the professor in advance so that we can arrange something else for you.



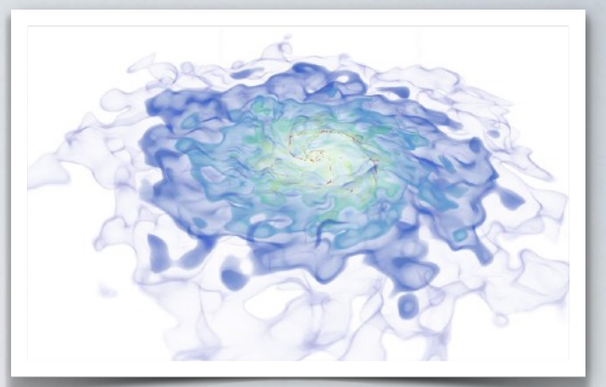
Midterm: On The Day of the Exam (II)

- Make sure to sit far enough from one another to allow enough personal space.

- If any issue with any of the questions, please let the TAs know in the first 30 minutes. You may leave the room after 30 minutes.



Seats in 56-105/106



One of the doors to Bldg 56 will be propped open between 19:00-19:30 on Oct. 24 (Sat).